Midterm review session

GOVT 6039 · Statistical and Computational Models · Fall 2026

Coefficient interpretation, maximum likelihood and Bayesian inference

Instructor: Daniel de Kadt · Session led by Georgy · **Student version**

The midterm. Tuesday, October 27, in class, 75 minutes. It covers Units 1 and 2 (Lectures 1–14). You may bring two pages of notes and no devices.

Contents. Part I covers how to interpret a coefficient in each model from the course. Part II compares maximum likelihood with Bayesian inference. Each part starts with a summary of the main idea and the tools, followed by exercises in the style of the exam. Appendix A is a one-page summary that you can use as one of your two pages of notes.

Using the handout. Each exercise begins with a setup, which gives the model, the data and the numbers you need, and a task. Try the exercises before reading the answers. Most of the calculations can be done by hand with the pocket table (Table 3) and the values in Appendix A.

Notation and abbreviations. Notation follows the lectures: $\eta_i = x_i^\top \beta$, $g\{\mathrm{E}[Y_i \mid x_i]\} = \eta_i$, $L(\theta; Y) = p(Y \mid \theta)$, $\hat{\pi}(x) = \text{logit}^{-1}(x^\top \hat{\beta})$. Lecture numbers appear in parentheses, so (L8) refers to Lecture 8. pp is percentage points, lgdp is the natural log of GDP per capita, OR is odds ratio, IRR is incidence rate ratio, PEA is the partial effect at the average, APE is the average partial effect, and ML and MLE are maximum likelihood and the maximum likelihood estimate.

Data used in the examples. Most numbers in this handout come from models estimated in lecture. The regime models (L7–L10, L12) use cross-national data with one row per country. The two covariates are oil, coded 1 for oil-producing states and 0 otherwise, and lgdp, the natural log of GDP per capita. One unit of lgdp is a 2.72-fold difference in income, and doubling income adds $\log 2 = 0.69$ to lgdp. The outcome differs by model: democracy coded 0/1 (LPM, logit and probit; L7–L8), regime type ordered from closed autocracy to electoral autocracy to democracy (ordered logit; L9), type of autocracy, which is military, single-party or personalist (multinomial logit; L9), the number of protests per year (Poisson; L10), and the number of years before a regime ends (discrete-time hazard; L12).

The Bayesian examples (L13–L14) use a survey of trust in local government in Chinese provinces, where $Y_i = 1$ if respondent i trusts local government. Hainan has $n = 15$ respondents, 12 of whom say yes, and Guangdong has $n = 236$. The parameter π is the share of all adults in the province who trust local government. Three examples are not from lecture and were chosen to keep the arithmetic simple: the turnout logit in Exercise 3, the baseline survival of 0.50 in Exercise 7 and the four-respondent province in Exercise 11.

Overview

Both review questions concern what an estimate from a fitted model means and how uncertain it is. Part I deals with the meaning of $\hat{\beta}$ in each model. Part II deals with two ways of describing uncertainty about an estimate. The box below states the main point of each part.

Part I. In every model in the course, a one-unit increase in x_{im} with the other covariates held fixed adds β_m to the linear predictor $\eta_i = x_i^\top \beta$. The models differ in how a change in η_i translates into a change in the outcome, and the link function g determines this. With the identity link, β_m is the change in the mean of Y. With any other link, the change in the outcome depends on the starting value of η_i, so it has to be computed as a predicted quantity.

Part II. Maximum likelihood and Bayesian inference use the same likelihood, $L(\theta; Y) = p(Y \mid \theta)$. ML reports the value of θ that maximizes L and measures uncertainty by how much $\hat{\theta}$ would vary across repeated samples. Bayesian inference multiplies L by a prior $p(\theta)$ and reports the posterior distribution $p(\theta \mid Y)$, which describes uncertainty about θ given the observed data. The prior is an assumption that has to be justified, and its influence declines as n grows. Posterior draws can be used to compute predicted quantities in the same way as the simulation method in Part I.

The plan for the 75-minute session is below. Material not covered in the session is written out in full in the handout.

Minutes	Sections	Content
0–5	Overview	main points of Parts I and II; interpretation steps (§1)
5–15	§1–2	Table 1 and Figure 1: odds ratios and changes in probability
15–45	§3	Exercises 1–8: interpretation and calculation
45–60	§4–6	ML and Bayesian inference for Hainan: Tables 4–5, Figures 3–4
60–72	§7–9	Exercises 9–15: priors, grid, Metropolis step, intervals
72–75	Wrap-up	summary sheet (Appendix A)

Part I Interpreting coefficients across models

1 Coefficients and the linear predictor

Every model in Units 1–2 has three parts (L7): a distribution for Y_i (normal, Bernoulli, Poisson and so on), a linear predictor η_i, and a link function g that connects the mean of Y_i to η_i,

$$\eta_i = x_i^\top \beta, \qquad g\big(\mathrm{E}[Y_i \mid x_i]\big) = \eta_i \quad \Longleftrightarrow \quad \mathrm{E}[Y_i \mid x_i] = g^{-1}(x_i^\top \beta).$$

A one-unit increase in x_{im}, with the other covariates fixed, changes η_i by β_m. Two consequences follow.

1. On the scale of η, the effect is the same for every unit and every value of x. This is why the log-odds lines in Figure 1 (left) are parallel and why the odds, rate and hazard ratios in Table 1 are constant. In the logit of democracy, oil lowers the log odds by 1.56 for poor and rich countries alike.
2. On the scale of $\mathrm{E}[Y_i \mid x_i] = g^{-1}(\eta_i)$, the effect equals β_m only when g is the identity. For other links g^{-1} is curved, so the same change in η produces a larger or smaller change in the outcome depending on the starting value of η_i, and therefore on all of x_i. The change of -1.56 in log odds lowers the probability of democracy by 8 percentage points for a country starting at $\pi = 0.10$ and by 33 points for a country starting at $\pi = 0.50$ (Table 2).

The link keeps $\mathrm{E}[Y_i \mid x_i]$ within the range allowed by the outcome. Probabilities must lie between 0 and 1 (logit, probit and cloglog links), and expected counts must be positive (log link). A linear function cannot satisfy these limits, so the model is linear in η and nonlinear on the scale of the outcome. Because the slope of a nonlinear function changes along the curve, a single coefficient implies different effects on the outcome for different units.

> **Steps for interpreting one coefficient** $\hat{\beta}_m$ (L8, L9, L10, L12). We interpret the estimate $\hat{\beta}$, not the estimand β, and hold the other covariates fixed.
>
> 1. Read the sign. Outside the simplest models the sign does not always carry over to the outcome. The exceptions are the middle categories of an ordered outcome and any category of a multinomial outcome.
> 2. Exponentiate. $e^{\hat{\beta}_m}$ multiplies the odds (logit and ordered logit), the expected count (Poisson and negative binomial) or the hazard (cloglog). A multiplier of 0.21 is a 79% reduction, and a multiplier of 1.88 is an 88% increase.
> 3. For a change of size c, the multiplier is $e^{c\hat{\beta}_m}$. Doubling GDP per capita corresponds to $c = \log 2$, so the multiplier is $e^{\hat{\beta}_m \log 2} = 2^{\hat{\beta}_m}$.
> 4. If the reader needs probabilities, counts or survival, compute a predicted quantity (§2).
>
> Steps 2 and 3 follow from the link. Exponentiating reverses the log, so an additive change in η becomes a multiplicative change in the odds, the expected count or the hazard. As L8 notes, other GLMs require different versions of steps 2 and 3, and even step 1 may fail.

Table 1 applies these steps to each model in the course. The second column gives what $\hat{\beta}_m$ changes, the third gives what $e^{\hat{\beta}_m}$ multiplies, and the fourth says whether a predicted quantity is needed to describe the effect on the outcome.

Table 1: Interpretation of a one-unit increase in x_{im} in each model. Lecture numbers in parentheses.

Model, link g	$\hat{\beta}_m$ is the change in…	$e^{\hat{\beta}_m}$ is…	Predicted quantity needed?
OLS (L2) identity	$\hat{\mu}$, in units of Y, the same at every x. FWL: the slope of residualized Y on residualized x_m.	—	No. $\hat{\beta}_m$ is already the change in $\mathrm{E}[Y \mid x]$. (It is the slope of the best linear predictor $\beta^\dagger$, not necessarily of the true curve.)
Linear probability (L7) identity	$P(Y_i = 1 \mid x_i)$, in probability points; $\times 100$ gives percentage points.	—	Rarely. A constant slope cannot hold near 0 and 1, and 27 of 205 fitted "probabilities" fall outside $[0, 1]$.
Logit (L7–8) logit	the log odds of $Y_i = 1$.	the odds ratio: multiplies the odds $\pi_i/(1-\pi_i)$.	Yes, for any statement about probability: $\hat{\pi}(x)$, a contrast $\hat{\tau}_j(x)$, a PEA or APE. Slope: $\hat{\beta}_j \hat{\pi}(1-\hat{\pi})$.
Probit (L7) Φ^{-1}	the index $x_i^\top \beta$ (in standard-deviation units of the latent normal).	no odds-ratio reading	Yes, always. $\hat{\beta}$ is on a different scale from the logit coefficients, so compare $\hat{\pi}$ rather than $\hat{\beta}$.
Ordered logit (L9) cumulative logit	the log odds of being above any cutpoint ζ_j (proportional odds). The latent distribution shifts right by $\hat{\beta}_m$, and the $\hat{\zeta}_j$ do not change.	multiplies the odds of being above any cutpoint.	Yes. $P(Y_i \le j \mid x_i) = \mathrm{logit}^{-1}(\zeta_j - x_i^\top \beta)$. Only the first and last categories must follow the sign, and the three APEs sum to 0.
Multinomial logit (L9) baseline-category, $\beta_1 = 0$	the log odds of category j *relative to the base* category.	multiplies $P(Y_i = j \mid x_i)/P(Y_i = 1 \mid x_i)$.	Yes. $\hat{\beta}_{jm}$ does not determine the change in $P(Y_i = j \mid x_i)$, and a negative coefficient can occur with a positive APE.
Poisson, NB (L10) log	the log expected count.	the incidence rate ratio: multiplies $\mathrm{E}[Y_i \mid x_i]$, for every unit.	For counts. The difference is $\hat{\mu}(x)(e^{\hat{\beta}_m} - 1)$, which grows with $\hat{\mu}$, while the ratio is constant.
Discrete-time hazard (L12) cloglog	$\log\{-\log(1-h_{ij})\}$, the same in every spell year j (proportional hazards).	the hazard ratio: multiplies the hazard rate λ_{ij} ($\approx h_{ij}$ when small).	Yes, for survival: $\hat{S}(t \mid x) = \prod_{j \le t}\{1 - \hat{h}_j(x)\}$. The hazard ratio is not a ratio of survival probabilities.
Hierarchical logit (L14) logit, intercept $\beta_{0j} = \beta_0 + \beta_{\mathrm{lgdp}}\mathrm{lgdp}_j + u_j$	the log odds with u_j held fixed, i.e. for respondents in the same province.	odds ratio, within a province.	Yes, for provincial levels: $\hat{\pi}_c = \mathrm{logit}^{-1}(x_c^\top \hat{\beta} + \hat{u}_j)$, then poststratify (MRP).
Ridge, lasso (L5) identity, penalized	the shrunk coefficient $\hat{\beta}_m^\lambda$: pulled toward 0, depends on λ and on the units of x_m; biased by design.	—	It is a coefficient for prediction and should not be read as the effect of x_m. A lasso estimate of zero means that x_m was not selected.

2 Predicted quantities

2.1 Odds ratios and changes in probability

In the class logit, $\hat{\beta}_{\mathrm{oil}} = -1.560$, so oil lowers the log odds of democracy by 1.56 for every country and multiplies the odds by $e^{\hat{\beta}_{\mathrm{oil}}} = 0.21$ for every country. The change in probability varies across countries (Figure 1, right; Table 2). At the three values of lgdp marked in Figure 1 it is -5, -35 and -17 percentage points. The left panel of Figure 1 shows the model on the log-odds scale, where oil shifts a straight line down by a constant amount. The right panel shows the probability scale, where the same shift is small in the tails of the curve and large in the middle.

The odds ratio is constant but the change in probability is not. In the middle row of Table 2, a country with $\hat{\pi} = 0.50$ has odds of $0.50/0.50 = 1.00$. Oil multiplies the odds by 0.21, so the new odds are 0.21 and the new probability is $0.21/1.21 = 0.17$, a decrease of 33 points. The same calculation gives a decrease of 8 points for a country starting at 0.10 and 25 points for a country starting at 0.90. The size of the change depends on the starting value of π_i, which depends on all of x_i (here, on lgdp). We do not observe π_i, because each country either is or is not a democracy, so it has to be estimated from the model. This is the reason for computing predicted quantities. The derivative gives the same result. $\partial\hat{\pi}(x)/\partial x_j = \hat{\beta}_j \hat{\pi}(x)\{1 - \hat{\pi}(x)\}$ is largest at $\hat{\pi} = 0.5$, where $\hat{\pi}(1-\hat{\pi}) = 0.25$ (for lgdp, $1.851 \times 0.25 = 0.46$ per unit), and approaches zero as $\hat{\pi}$ approaches 0 or 1. It follows that a logit coefficient divided by 4 is the largest possible change in probability per unit of x_j, and this maximum applies only to units near $\hat{\pi} = 0.5$.

2.2 Choosing what to report (L8)

A predicted quantity is a value implied by the model on the scale of the outcome, such as a probability, an expected count or a survival probability, computed from $\hat{\beta}$ at chosen values of x. Reporting one involves three choices, and each should be stated.

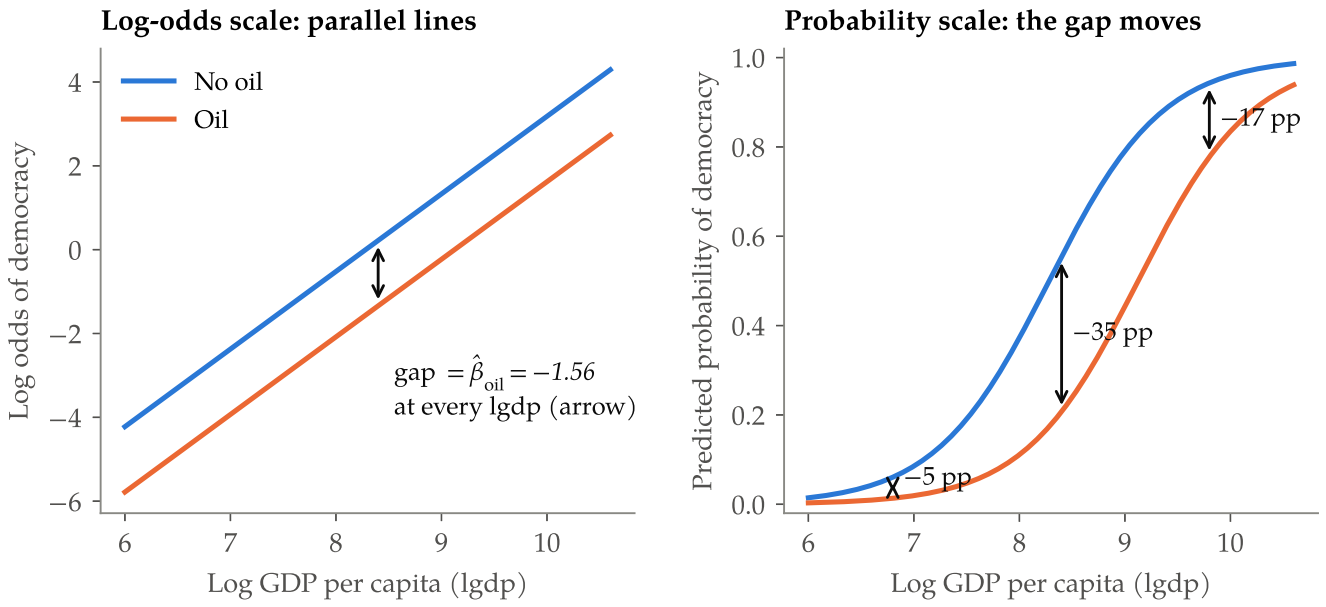


Figure 1: Logit of regime type on oil and lgdp, with the class estimates ($\hat{\beta}_0 = -15.329$, $\hat{\beta}_{\text{oil}} = -1.560$, $\hat{\beta}_{\text{lgdp}} = 1.851$; L8). The gap between the curves with and without oil is constant on the log-odds scale and varies on the probability scale.

Table 2: Switching oil on multiplies every country's odds by $e^{\hat{\beta}_{\text{oil}}} = 0.21$ (L8). By hand, new odds = old odds $\times 0.21$ and $\pi = \text{odds}/(1 + \text{odds})$.

$\hat{\pi}$ without oil	odds	odds with oil	$\hat{\pi}$ with oil	change (pp)
0.1	0.11	0.02	0.02	−8
0.5	1.00	0.21	0.17	−33
0.9	9.00	1.89	0.65	−25

Contrast. $\hat{\tau}_j(x) = \hat{\pi}(x_j = a, x_{-j}) - \hat{\pi}(x_j = b, x_{-j})$ is the difference between the predicted probability with $x_j = a$ and with $x_j = b$, other covariates unchanged. For a binary x_j, use $a = 1$ and $b = 0$. For a continuous x_j, choose a change that is easy to describe, such as one unit of lgdp (a 2.72-fold increase in GDP per capita), a doubling of GDP per capita ($c = \log 2$), one standard deviation, or the interquartile range. The derivative is the limit as the change goes to zero. Different contrasts give different numbers (21.7, 16.0, 40.4 and 49.7 points in L8), so the contrast should always be named.

Values of the other covariates. The contrast depends on the other covariates. The PEA (partial effect at the average) holds them at typical values, $\hat{\tau}^{\text{PEA}} = \hat{\pi}(1, \overline{\text{lgdp}}) - \hat{\pi}(0, \overline{\text{lgdp}})$ (L8: −0.366). It describes a single hypothetical country with average characteristics. The APE (average partial effect) computes the contrast for each country at its observed x_i and averages the results, $\hat{\tau}^{\text{APE}} = \frac{1}{n}\sum_i \hat{\tau}_i$ (−0.229). It describes the countries in the data. Because the model is nonlinear, the effect at the average country is generally different from the average effect. Hanmer and Kalkan (2013) recommend the observed-value approach (APE).

Uncertainty. The reported quantity $T(\hat{\beta})$ is a function of $\hat{\beta}$ and inherits its uncertainty. Since $\hat{\beta} \overset{\text{approx.}}{\sim} N\big(\beta, [-H(\hat{\beta})]^{-1}\big)$, draw $\beta^{(k)} \sim N\big(\hat{\beta}, \widehat{\text{Var}}(\hat{\beta})\big)$ for $k = 1, \ldots, R$, compute $T(\beta^{(k)})$, and report the standard deviation and the 2.5th and 97.5th percentiles of the draws. The delta method is an alternative (L8). Simulation is used because T is a nonlinear function of $\hat{\beta}$, so its standard error is not a simple multiple of $\text{se}(\hat{\beta})$. The spread of T across draws of β gives its uncertainty without further algebra.

Elements of a complete interpretation. The midterm rewards interpretations that state (i) the quantity and the contrast, (ii) the size of the change in x, (iii) the values at which the other covariates are held (PEA or APE), (iv) the scale, and (v) the uncertainty.

An example: "Holding lgdp at each country's observed value, switching oil from 0 to 1 lowers the predicted

> probability of democracy by 23 percentage points on average (APE -0.229, 95% interval $[-0.346, -0.111]$; L8)." The quantity and contrast are $\hat{\pi}$ for oil equal to 1 versus 0, the change is switching oil on, the covariates are held at observed values, the scale is percentage points, and the uncertainty is given by the interval.

3 Exercises for Part I

Exercise 1 asks for interpretations in words, Exercises 2–7 ask for predicted quantities computed by hand, and Exercise 8 asks you to find errors in interpretation. The instructor version shows each step of the answers.

Exercise 1. Interpreting coefficients in words

Setup. Each row gives a coefficient from a model estimated in lecture, along with the outcome and covariates. Use the steps in §1: read the sign, exponentiate if the link is not the identity, and state what the multiplier applies to. A complete sentence states what is held fixed, the size of the change in x and the scale, and uses units that a reader understands, such as percentage points or "times as many".
Task. Write one sentence for each row. For the log-scale rows, the values of $e^{0.5}$, e^{1} and e^{2} in the caption of Table 3 can be used to check the size of each multiplier.

Model, outcome and covariates	Output	One sentence
OLS (L2): turnout (pp) on age and mail	$\hat{\beta}_{\text{age}} = 0.585$	
LPM (L7): democracy (0/1) on oil and lgdp	$\hat{\beta}_{\text{oil}} = -0.22$	
Logit (L8): democracy on oil and lgdp	$\hat{\beta}_{\text{oil}} = -1.560$	
Ordered logit (L9): closed autocracy, electoral autocracy, democracy, on lgdp and oil	$\hat{\beta}_{\text{lgdp}} = 1.96$	
Multinomial logit (L9): military, single-party or personalist, on lgdp and oil; base category military	$\hat{\beta}_{\text{pers,oil}} = -2.37$	
Poisson (L10): protests per year on lgdp, e_autocracy and oil	$\hat{\beta}_{\text{e_autocracy}} = 0.63$	
Cloglog hazard (L12): yearly probability that a regime ends, on lgdp and oil	$\hat{\beta}_{\text{lgdp}} = -0.73$	

Exercise 2. The class logit (L8)

Setup. The class logit of democracy ($Y_i = 1$) on oil and lgdp has $\hat{\beta}_0 = -15.329$, $\hat{\beta}_{\text{oil}} = -1.560$ and $\hat{\beta}_{\text{lgdp}} = 1.851$. A country's linear predictor is $\hat{\eta} = -15.329 - 1.560\,\text{oil} + 1.851\,\text{lgdp}$, its predicted odds are $e^{\hat{\eta}}$, and its predicted probability is $\hat{\pi} = e^{\hat{\eta}}/(1+e^{\hat{\eta}}) = 1/(1+e^{-\hat{\eta}})$. Doubling income adds $\log 2 = 0.69$ to lgdp.
Task. (a) Interpret $\hat{\beta}_{\text{oil}}$ on the log-odds scale and as an odds ratio. Why is neither enough for a reader? (b) By what factor does doubling GDP per capita multiply the odds? (c) For a country with lgdp $= 8.5$, compute $\hat{\pi}$ with and without oil, and the change. (d) The slides report $\hat{\tau}^{\text{PEA}} = -0.366$ and $\hat{\tau}^{\text{APE}} = -0.229$. Why do they differ, and which would you report?

Exercise 3. Logit predictions by hand

Setup. A hypothetical logit of turnout ($Y_i = 1$ if person i votes) has $\hat{\eta} = -3 + 1 \cdot$ decades since age $18 + 2 \cdot$ contact, where contact equals 1 if the person received a mobilization call and 0 otherwise. Ages 18, 48 and 68 correspond to 0, 3 and 5 decades since age 18. All values of $\hat{\eta}$ here are integers, so each $\hat{\pi}$ can be read from Table 3.
Task. (a) What is the odds ratio for contact? (b) Find $\hat{\pi}$ with and without contact at ages 18, 48 and 68, and the change at each age. (c) What single number would you report for the effect of contact, and what other information is needed?

Table 3: Pocket table: $\text{logit}^{-1}(z) = e^z/(1+e^z)$. Also useful: $e^{0.5} = 1.65$, $e^1 = 2.72$, $e^2 = 7.39$, $e^{-1} = 0.37$, $e^{-2} = 0.14$, $\log 2 = 0.69$.

z	-4	-3	-2	-1	0	1	2	3	4
$\text{logit}^{-1}(z)$	0.018	0.047	0.119	0.269	0.500	0.731	0.881	0.953	0.982

Exercise 4. Ordered logit (L9)

Setup. The outcome has three ordered categories: closed autocracy (1), electoral autocracy (2) and democracy (3). The model assumes a latent variable $Y_i^* = x_i^\top \beta + \varepsilon_i$ with a logistic error. A country is a closed autocracy if Y_i^* is below the first cutpoint ζ_1, an electoral autocracy if it is between ζ_1 and ζ_2, and a democracy if it is above ζ_2. The cutpoints replace the intercept. The class estimates are $\hat{\beta}_{\text{lgdp}} = 1.96$, $\hat{\beta}_{\text{oil}} = -1.54$, $\hat{\zeta}_1 = 13.75$ and $\hat{\zeta}_2 = 16.30$.
Formulas. $P(\text{closed}) = \text{logit}^{-1}(\hat{\zeta}_1 - x_i^\top \hat{\beta})$ and $P(\text{closed or electoral}) = \text{logit}^{-1}(\hat{\zeta}_2 - x_i^\top \hat{\beta})$. $P(\text{electoral})$ is the difference between these two, and $P(\text{democracy}) = 1 - \text{logit}^{-1}(\hat{\zeta}_2 - x_i^\top \hat{\beta})$.
Task. Complete the table for countries with lgdp of 8.5 and 6.0, with and without oil. Then answer the following. (a) Which categories change in the direction implied by the sign of $\hat{\beta}_{\text{oil}}$? (b) Why do the three changes sum to zero? (c) At approximately what value of lgdp does the effect of oil on electoral autocracy change sign?

lgdp	oil	$x_i^\top \hat{\beta}$	$\hat{\zeta}_1 - x_i^\top \hat{\beta}$	$\hat{\zeta}_2 - x_i^\top \hat{\beta}$	$\hat{P}$(closed)	$\hat{P}$(electoral)	$\hat{P}$(democracy)
8.5	0						
8.5	1						
change from switching oil on							
6.0	0						
6.0	1						
change from switching oil on							

Exercise 5. Multinomial logit (L9)

Setup. The outcome is the type of autocracy, with three unordered categories: military (the base category), single-party and personalist. Each non-base category j has its own coefficient vector β_j. For the base category $\beta_1 = 0$, so its score is always $e^0 = 1$. The probability of a category is its score divided by the sum of the scores, $P(Y_i = j \mid x_i) = \exp(x_i^\top \beta_j) / \sum_k \exp(x_i^\top \beta_k)$. The class estimates for (intercept, lgdp, oil) are $\hat{\beta}_{\text{sp}} = (-12.78,\ 1.75,\ -0.94)$ for single-party and $\hat{\beta}_{\text{pers}} = (-25.68,\ 3.34,\ -2.37)$ for personalist.

Task. For lgdp of 8, compute the scores $\exp(x_i^\top \hat{\beta}_j)$ and the probabilities with and without oil. Then use these numbers to explain why oil raises the probability of single-party rule although its single-party coefficient is negative.

	score $\exp(x_i^\top \hat{\beta}_j)$			probability		
	military	single-party	personalist	military	single-party	personalist
oil $= 0$						
oil $= 1$						
change						

Exercise 6. Poisson regression (L10)

Setup. The outcome is the number of protests in a country per year. The Poisson model uses a log link, so the coefficients act on the log of the expected count μ_i. The class estimates are $\log \hat{\mu}_i = -2.10 + 0.30\,\text{lgdp}_i + 0.63\,\text{e_autocracy}_i - 0.03\,\text{oil}_i$, where e_autocracy equals 1 for electoral autocracies and 0 otherwise. A student writes: "electoral autocracies have 0.63 more protests."

Task. (a) What is wrong with this statement? (b) Give the correct interpretation. (c) For oil $= 0$, compute $\hat{\mu}$ for e_autocracy $= 0$ and 1 at lgdp $= 8$ and lgdp $= 10$, and the difference in each case. (d) What would you report if asked how many more protests electoral autocracies have?

Exercise 7. Discrete-time hazard model (L12)

Setup. Each regime is observed year by year. The hazard h_{ij} is the probability that regime i ends in spell year j, given that it lasted until year j. The model uses a complementary log-log (cloglog) link, $\log\{-\log(1 - h_{ij})\} = \gamma_j + x_i^\top \beta$, where γ_j is the baseline hazard for spell year j. Survival to year t is the probability that the regime does not end in any year up to t, $S(t \mid x) = \prod_{j \le t}(1 - h_{ij})$. The class estimates are $\hat{\beta}_{\text{lgdp}} = -0.73$ and $\hat{\beta}_{\text{oil}} = -0.53$.

Formulas. From L12, $1 - h_{ij} = e^{-\lambda_{ij}}$, where $\lambda_{ij} = e^{\gamma_j + x_i^\top \beta}$ is the hazard rate.

Task. (a) Interpret each coefficient as a hazard ratio. (b) A regime without oil has $\hat{S}(20 \mid x) = 0.50$. What is $\hat{S}(20 \mid x)$ for an otherwise identical regime with oil? (c) Why is the answer not 0.50×0.59 or $0.50/0.59$?

Exercise 8. Errors in interpretation

Setup. Each sentence below could appear in an exam answer and contains one error, usually about the scale (log odds, odds or probability) or about the comparison (relative to the base category, or to the hazard rather than survival).
Task. Explain the error and write a corrected sentence.

	Sentence	Error and correction
1	"Oil lowers the probability of democracy by 1.56." (logit)	
2	"Oil states are 0.21 times as likely to be democracies."	
3	"Electoral autocracies have 1.88 more protests."	
4	"Oil's coefficient on single-party is -0.94, so oil lowers $P(\text{single-party})$." (multinomial)	
5	"Oil's coefficient is negative, so oil lowers every higher category." (ordered)	
6	"A hazard ratio of 0.59 means oil regimes are 0.59 times as likely to survive 20 years."	
7	"The logit coefficient (1.851) exceeds the probit's, so logit finds a bigger effect."	
8	"The lasso set $\hat{\beta}_m = 0$, so x_m has no effect."	

Part II Maximum likelihood versus Bayesian inference

4 The likelihood

Both approaches start from a probability model for the data and from the data themselves. Here the model assumes that each respondent independently says yes with the same probability, $Y_i \sim \text{Bernoulli}(\pi)$. The likelihood (L6) is the probability of the observed data, written as a function of π,

$$L(\pi; Y) = p(Y \mid \pi) = \prod_{i=1}^{n} \pi^{Y_i}(1-\pi)^{1-Y_i} = \pi^{r}(1-\pi)^{n-r}, \qquad \ell(\pi) = r \log \pi + (n-r)\log(1-\pi),$$

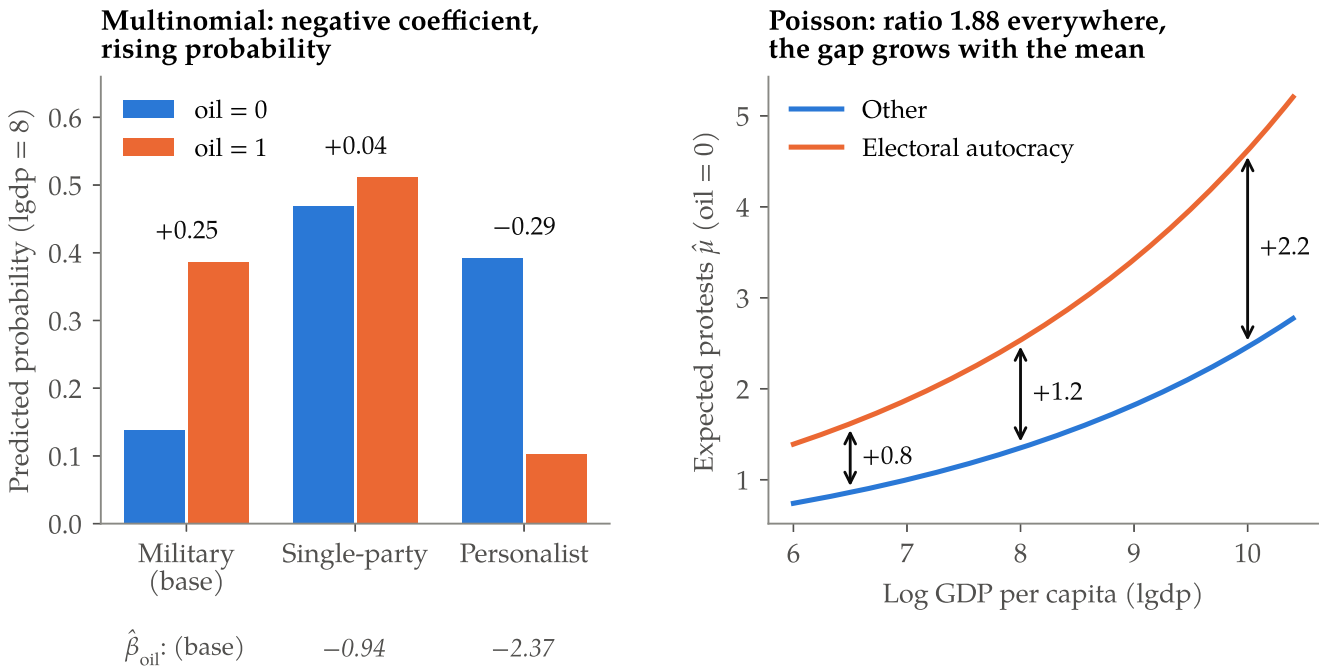


Figure 2: Left (Exercise 5): multinomial probabilities at lgdp = 8, military as base. Right (Exercise 6): Poisson expected protests at oil = 0, from the two-decimal coefficients printed on the L10 slide (the slide's own figure uses unrounded estimates, so its gaps differ slightly).

where $r = \sum_i Y_i$ is the number of yes answers. In the running example (L13), $n = 15$ respondents in Hainan, $r = 12$ of them trust local government, and π is the share of all adults in Hainan who do. In both approaches π is fixed and unknown (L13). The approaches differ in what they treat as random and therefore in how they define uncertainty. Table 4 compares them using the same data.

ML asks which value of π makes the observed data most likely and then asks how much this estimate would vary across repeated surveys. Bayesian inference asks how plausible each value of π is, given the prior and the observed data. The ML standard error comes from the curvature of the log likelihood at its maximum, $-H(\hat{\pi}) = r/\hat{\pi}^2 + (n-r)/(1-\hat{\pi})^2 = n/\{\hat{\pi}(1-\hat{\pi})\} = 15/(0.80 \times 0.20) = 93.75$, so $\text{se} = 1/\sqrt{93.75} = 0.103$. A more sharply curved log likelihood gives a smaller standard error. For Hainan, the Wald interval $[0.598, 1.002]$ extends past 1 because the normal approximation is poor with $n = 15$ and $\hat{\pi}$ close to 1. The posterior stays within $[0, 1]$ by construction. This is a limitation of the normal approximation and not of ML in general.

5 Prior, likelihood and posterior

The prior is $\pi \sim \text{Beta}(a, b)$, with $\text{E}[\pi] = a/(a+b)$. The Beta distribution is used because it is defined on $[0, 1]$, like π, and its parameters can be read as prior counts of yes and no answers. Matching the sample shares in the other 30 provinces (mean 0.58, sd 0.15) gives $\text{Beta}(6, 4)$, with mean 0.60, sd 0.15 and a "worth" of $a + b = 10$ respondents. The likelihood is $L(\pi; Y) = \pi^{12}(1-\pi)^3$. Bayes' rule combines the two:

$$\underbrace{p(\pi \mid Y)}_{\text{posterior}} = \frac{\overbrace{p(Y \mid \pi)}^{\text{likelihood } L(\pi;Y)} \; \overbrace{p(\pi)}^{\text{prior}}}{\underbrace{p(Y)}_{\text{normalizing constant}}} \propto L(\pi; Y)\, p(\pi).$$

The term $p(Y)$ does not depend on π and only makes the posterior integrate to one, so it can be dropped when we are interested in the shape of the posterior. With a Bernoulli likelihood the Beta prior is conjugate, which means that the posterior is also a Beta distribution. Multiplying $\pi^{a-1}(1-\pi)^{b-1}$ by $\pi^r(1-\pi)^{n-r}$ adds the exponents, so

$$\pi \mid Y \sim \text{Beta}(a + r,\; b + n - r) = \text{Beta}(18, 7),$$

Table 4: The two approaches on the same data (Hainan, $r = 12$, $n = 15$). Numbers from L6, L8 and L13.

	Maximum likelihood	Bayesian inference
Ingredients	model for Y only	model for Y *and* a prior $p(\pi)$
The likelihood	$L(\pi; Y)$: a function of π, not a distribution for it	the same $L(\pi; Y)$, now multiplied by the prior
Target	the π that maximizes ℓ: $\hat{\pi} = r/n = 0.80$	a distribution, the posterior $p(\pi \mid Y) \propto L(\pi; Y)\, p(\pi)$
What is random	the data, over repeated samples, so $\hat{\pi}$ varies from sample to sample	π is fixed, and $p(\pi \mid Y)$ describes uncertainty about it given the observed data
Uncertainty	$\text{se}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/n} = 0.103$, from the curvature $-H(\hat{\pi}) = 93.75$ (observed information)	the spread of the posterior: sd $= 0.088$
Interval	95% **confidence** interval $\hat{\pi} \pm 1.96\,\text{se} = [0.598, 1.002]$. The 95% describes the procedure over repeated samples	95% **credible** interval [0.53, 0.87]: given our model and prior, a 95% probability that π lies in it
Probability of a claim	none, since π is not random and "$P(\pi > 0.6)$" has no meaning	$P(\pi > 0.6 \mid Y) = 0.90$
Computation	solve the score $s(\hat{\theta}) = 0$: closed form, Newton–Raphson (Fisher scoring in `glm()`), `optim()`	closed form if the prior is conjugate; otherwise grid, Metropolis/MCMC, Stan (HMC)
Assumes	the model is true; a random sample; n large for $\hat{\beta} \overset{\text{approx.}}{\sim} N$	the model and the prior, an additional assumption that is stated explicitly (L13)

Table 5: Posterior means under four priors (L13). "Worth" is $a + b$. A prior with a larger worth is more concentrated and carries more information.

Prior	Worth	Prior sd	Hainan ($n = 15$)	Guangdong ($n = 236$)
None (MLE)			0.800	0.466
Beta$(1, 1)$	2	0.29	0.765	0.466
Beta$(6, 4)$	10	0.15	0.720	0.472
Beta$(12, 8)$	20	0.11	0.686	0.477

$$\text{E}[\pi \mid Y] = \underbrace{\tfrac{a+b}{a+b+n}}_{\text{weight on prior}} \cdot \tfrac{a}{a+b} + \underbrace{\tfrac{n}{a+b+n}}_{\text{weight on data}} \cdot \tfrac{r}{n} = 0.40 \cdot 0.60 + 0.60 \cdot 0.80 = 0.72.$$

The posterior mean is a weighted average of the prior mean and the MLE. The weights depend only on the prior worth, $a + b = 10$, and the sample size, $n = 15$, so the data receive $15/25 = 60\%$ of the weight. As n increases, the weight on the data approaches 100% and the prior has less influence.

The posterior lies between the prior and the likelihood, closer to whichever carries more information. For Hainan the prior receives 40% of the weight and moves the estimate by up to 11 points (under the strongest prior in Table 5). For Guangdong it receives 4% of the weight and moves the estimate by at most 1 point. In Figure 3 the Hainan posterior lies clearly between the prior and the likelihood, while the Guangdong posterior nearly coincides with the likelihood. The prior has the most influence when the sample is small, which is also when its source most needs to be justified.

> **Summary of the components.** The prior $p(\pi)$ describes which values of π are considered plausible before seeing Y. It is an assumption, which can be summarized as pseudo-data worth $a + b$ respondents, and it does not imply that π is random. The likelihood $L(\pi; Y)$ measures how well each value of π accounts for the observed data. It is not a probability distribution for π, since its integral is not one, and it is the only contribution of the data in both approaches. The posterior $p(\pi \mid Y)$ is the product of prior and likelihood, rescaled to integrate to one, and describes uncertainty about π after observing the data. It can be summarized by its mean, a credible interval or a probability such as $P(\pi > 0.6 \mid Y)$. Under ML, the uncertainty of an estimate is the sampling variability of $\hat{\pi}$ (standard error and confidence interval). Under Bayesian inference it is the spread of the posterior (standard deviation and credible interval).

6 Sampling variability and posterior uncertainty

Figure 4 compares the two. The left panel shows the ML view. With π fixed at 0.67 (the value used in the L13 simulation), 20,000 surveys of $n = 15$ are drawn and $\hat{\pi}$ is computed for each. The standard deviation of these estimates is about 0.12, as in the L13 table, and this sampling variability is the basis of a confidence interval. The observed $\hat{\pi} = 0.80$ is one draw from this distribution, and a different sample of 15 respondents would give a different estimate. The right panel shows the Bayesian view. It uses only the observed survey and describes uncertainty about

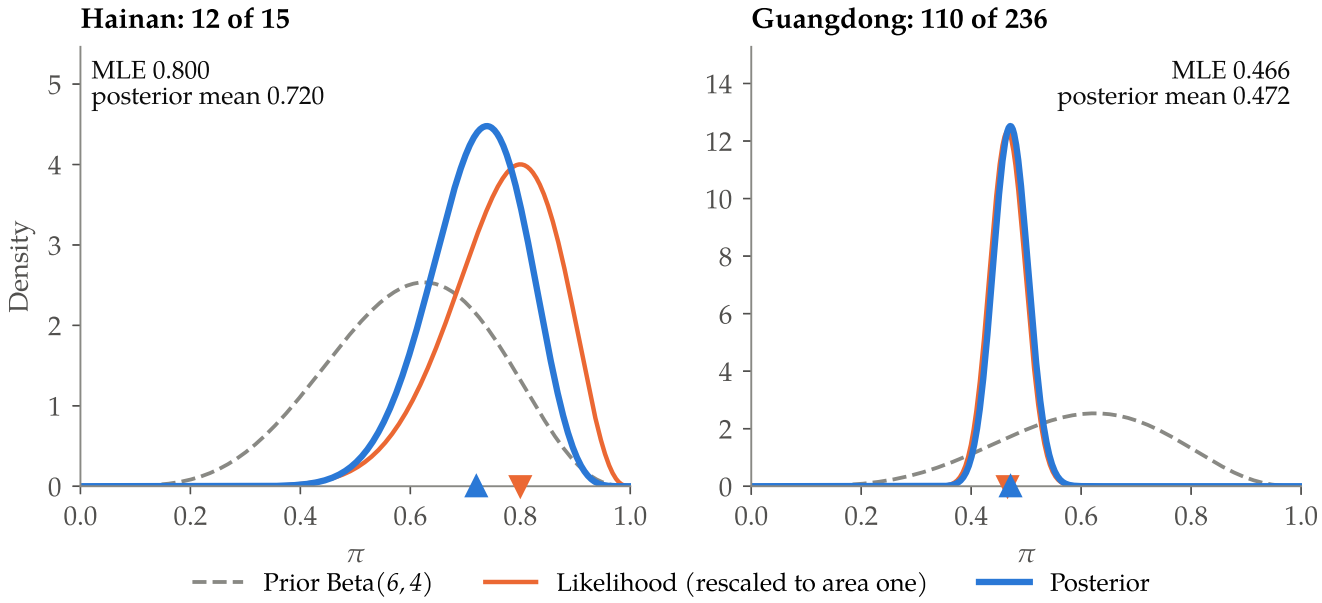


Figure 3: Prior, rescaled likelihood and posterior for two provinces under the same prior, Beta$(6, 4)$. Triangles mark the MLE (orange, pointing down) and the posterior mean (blue, pointing up). The prior moves the estimate by 8 points for Hainan ($n = 15$) and by less than one point for Guangdong ($n = 236$).

the fixed π with the posterior, without reference to repeated surveys.

> **Correct statements.** For the confidence interval: "If the survey were repeated many times, 95% of the intervals constructed in this way would contain π." For the credible interval: "Given the model and the prior, the probability that π lies in $[0.53, 0.87]$ is 0.95."
>
> **Incorrect statement.** For a single confidence interval: "The probability that π lies in $[0.598, 1.002]$ is 0.95." Once the interval is computed, it either contains the fixed π or it does not. The 95% refers to the procedure and not to a particular interval.

7 Priors on regression coefficients

In regression models the priors are placed on the coefficients, for example $\beta_j \sim N(m_j, s_j^2)$. A normal prior is used instead of a Beta prior because coefficients can take any real value (L13). The prior mean m_j is the expected value of the coefficient, and the prior standard deviation s_j reflects how confident we are in it.

One reason to use a prior is separation. In the Hainan logit only one respondent is aged 60 or older, and she trusts local government. Increasing β_{60+} moves her predicted probability closer to 1 and always increases the likelihood, so the likelihood has no maximum. The MLE is $+\infty$. The `glm()` function stops at 16 with a standard error of 3,956 and predicts Pr(trust) $= 1.00$ based on one respondent. With a prior that gives little weight to extreme values, the product $L \times p$ has a finite maximum, because the likelihood levels off as β_{60+} increases while the prior density continues to fall.

Prior on β_{60+}	$\hat{\beta}_{60+}$ (truth 0.55)	Pr(trust), urban, 60+
None (`glm()`)	16.0 (se 3,956)	1.00
Weak, $N(0, 2.5^2)$	0.95 [−3.00, 5.43]	0.84 [0.24, 1.00]
From the other provinces, $N(0.55, 0.5^2)$	0.64 [−0.32, 1.65]	0.75 [0.45, 0.93]

Posterior probabilities depend on the prior. L13 lists four ways to limit or justify its influence. The first is to have enough data that the prior receives little weight, as in Guangdong. The second is to check sensitivity by reporting results under several priors. The third is to use weak priors. A $N(0, 2.5^2)$ prior on a logit coefficient excludes only odds ratios above about $e^5 \approx 148$. The fourth is to estimate the prior with a hierarchical model, $\beta_{0j} = \mu + u_j$, $u_j \sim N(0, \sigma_u^2)$,

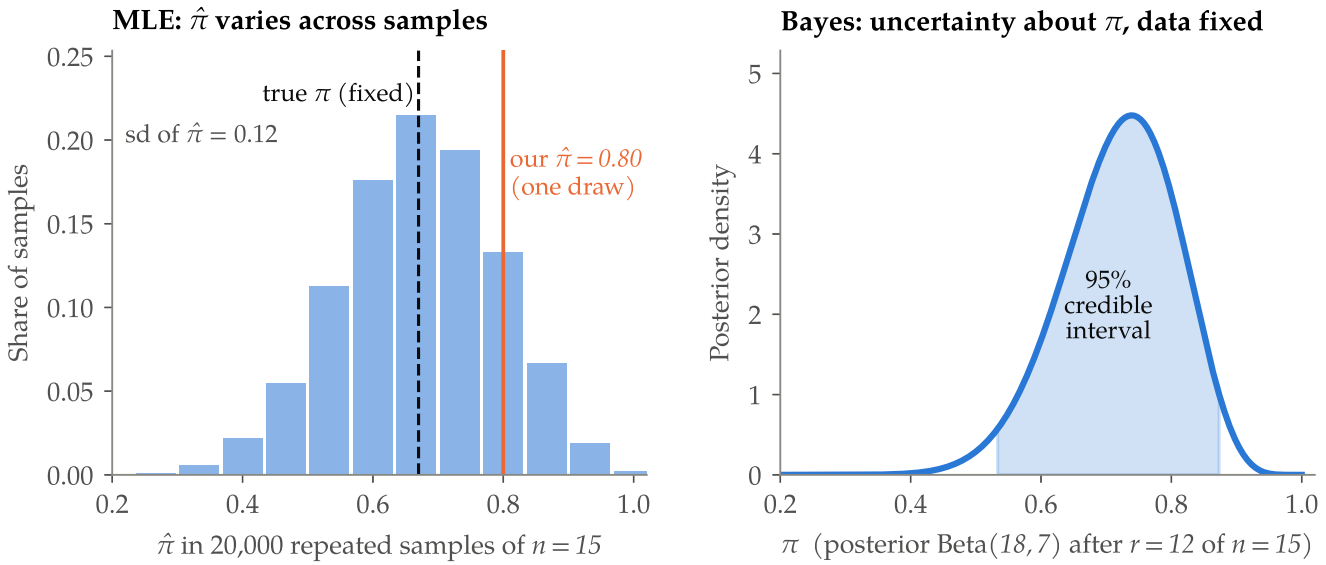


Figure 4: Left: sampling distribution of $\hat{\pi}$ across repeated surveys of $n = 15$ with $\pi = 0.67$. Right: posterior Beta$(18, 7)$ with its 95% credible interval.

which estimates from the data how similar the provinces are (empirical Bayes and partial pooling; L14). In such a model a small province such as Hainan is shrunk toward the national average by an amount estimated from the data rather than chosen by the analyst.

> **Connection to Part I.** In a Bayesian regression with a weak prior, β_m has the same interpretation as before. It is a change in log odds, and e^{β_m} is an odds ratio. The difference is that β now has a posterior distribution. The posterior draws $\beta^{(k)}$ take the place of the draws from $N(\hat{\beta}, \widehat{\text{Var}}(\hat{\beta}))$ in the L8 simulation method. For each draw, compute $T(\beta^{(k)})$, for example a predicted probability or an APE, and summarize the results. Part I covers what an estimate means and Part II covers how to describe its uncertainty. In both cases the reported quantity can be computed from draws of β.

8 Approximating the posterior

Conjugate priors are the exception. For the logit no standard prior is conjugate, and the normalizing constant $p(Y) = \int L(\theta; Y)\, p(\theta)\, d\theta$, an integral over all values of all coefficients, has no closed form. The posterior therefore has to be approximated.

A grid approximation evaluates prior times likelihood at many values and rescales the results to sum to one (Exercise 10 uses five values). The number of evaluations grows quickly with the number of parameters. One coefficient at 100 values requires 100 evaluations, and four coefficients require 10^8.

The Metropolis algorithm (L13) instead generates a sequence of values from the posterior.

1. From the current value $\pi^{(t)}$, propose $\pi^* = \pi^{(t)} + \varepsilon$ with $\varepsilon \sim N(0, s^2)$.
2. Compute ratio $= \dfrac{L(\pi^*; Y)\, p(\pi^*)}{L(\pi^{(t)}; Y)\, p(\pi^{(t)})}$.
3. Move to π^* with probability $\min(1, \text{ratio})$. Otherwise stay at $\pi^{(t)}$ and record it again.

Because $p(Y)$ appears in both the numerator and the denominator, it cancels from the ratio and never has to be computed. Proposals that increase the posterior (ratio above 1) are always accepted, and proposals that decrease it are accepted only some of the time. The chain therefore spends more time where the posterior is high, and the recorded values are draws from the posterior. In practice, run four chains from different starting values, discard the burn-in (the early draws before the chain reaches the posterior) and check $\hat{R}$, which compares the chains. Values below 1.05 suggest convergence. Stan uses Hamiltonian Monte Carlo, which uses the gradient of the log posterior to make longer moves.

9 Exercises for Part II

Unless stated otherwise, these exercises use the Hainan data ($r = 12$, $n = 15$), the Beta$(6, 4)$ prior and the Beta$(18, 7)$ posterior.

Exercise 9. Does the likelihood integrate to one?

Setup. For Hainan, $L(\pi; Y) = \pi^{12}(1-\pi)^3$. A probability density for π must integrate to one over $[0, 1]$. Figure 3 includes a curve labelled "likelihood (rescaled to area one)".

Formulas. For integers a and b, $\int_0^1 \pi^{a-1}(1-\pi)^{b-1}\, d\pi = B(a, b) = \dfrac{(a-1)!\,(b-1)!}{(a+b-1)!}$.

Task. Is $\int_0^1 L(\pi; Y)\, d\pi = 1$? Compute the integral and explain what the rescaled curve in Figure 3 is.

Exercise 10. A five-point grid (L13)

Setup. Suppose π can take only the five values 0.5, 0.6, 0.7, 0.8 and 0.9. The prior column gives the Beta$(6, 4)$ density at each value, rescaled to sum to one, and $L(\pi; Y) = \pi^{12}(1-\pi)^3$ for Hainan.

Formulas. On a grid, Bayes' rule has three steps. Multiply the prior by the likelihood in each row, sum the products to obtain $p(Y)$ for the grid, and divide each product by the sum.

Task. Complete the columns "Prior $\times L$" and "Posterior". Then compute the posterior under a flat prior (0.2 at each value). Where does each posterior reach its maximum, and why do the two differ?

π	$L(\pi; Y)$	Prior	Prior $\times L$	Posterior	Posterior, flat prior
0.5	3.1×10^{-5}	0.23			
0.6	1.4×10^{-4}	0.30			
0.7	3.7×10^{-4}	0.27			
0.8	5.5×10^{-4}	0.16			
0.9	2.8×10^{-4}	0.04			
Sum		1.00			

Exercise 11. A conjugate update by hand

Setup. A small province has $n = 4$ respondents, and $r = 1$ of them trusts local government, so the MLE is $1/4 = 0.25$. Use the same Beta$(6, 4)$ prior as for Hainan (mean 0.60, worth 10 respondents).

Formulas. $\pi \mid Y \sim \text{Beta}(a + r,\ b + n - r)$, and the posterior mean written as a weighted average (§5).

Task. (a) What is the posterior distribution? (b) Write the posterior mean as a weighted average of the prior mean and the MLE. (c) Which receives more weight, and what follows from this?

Exercise 12. Confidence and credible intervals

Setup. Hainan has two 95% intervals for π: the Wald confidence interval $[0.598, 1.002]$ from ML and the credible interval $[0.53, 0.87]$ from the Beta$(18, 7)$ posterior. Under ML the data are random across repeated surveys. Under Bayesian inference π is fixed, and uncertainty about it is described by a probability distribution.
Task. For each statement, indicate whether it correctly describes each interval (Yes or No), and explain.

	Statement	Confidence interval	Credible interval
a	"Over repeated surveys, 95% of intervals built this way cover π."		
b	"Given our model and prior, π lies in the interval with probability 0.95."		
c	"The probability that more than 60% of Hainan adults trust is 0.90."		
d	"The interval can extend beyond $[0, 1]$."		

Exercise 13. One Metropolis step

Setup. Hainan, Beta$(6, 4)$ prior, $r = 12$, $n = 15$. A Metropolis chain is at $\pi^{(t)} = 0.60$. The algorithm proposes a value π^*, computes the ratio of prior times likelihood at π^* to its value at $\pi^{(t)}$, and moves with probability $\min(1, \text{ratio})$ (§8).
Formulas. A Beta(a, b) density is proportional to $\pi^{a-1}(1-\pi)^{b-1}$. Also, $\log(7/6) = 0.154$, $\log(3/4) = -0.288$ and $\log(2/3) = -0.405$.
Task. (a) Show that $L(\pi; Y)\, p(\pi) \propto \pi^{17}(1-\pi)^{6}$. (b) What happens if the proposal is $\pi^* = 0.70$, and what happens if it is $\pi^* = 0.40$? Decide first without a calculator, then compute the ratios. (c) Why is $p(Y)$ not needed?

Exercise 14. Priors on coefficients

Setup. This exercise returns to the separation problem in §7. One respondent in Hainan is aged 60 or older and she trusts local government, so the MLE of β_{60+} is $+\infty$. Two priors are considered, a weak $N(0, 2.5^2)$ prior and an informative $N(0.55, 0.5^2)$ prior based on the other provinces. The results are in the table in §7.
Task. (a) What does a $N(0, 2.5^2)$ prior on a logit coefficient rule out? (b) Why is the posterior for β_{60+} finite even though only one respondent is 60 or older, and why is its interval almost as wide as the prior's? What does the informative prior gain, and at what cost?

Exercise 15. Errors in interpretation (Part II)

Setup. As in Exercise 8, each sentence contains one error, here about the likelihood, the prior or an interval.
Task. Explain the error and write a corrected sentence.

	Sentence	Error and correction
1	"The posterior is the likelihood."	
2	"ML is objective; Bayes is subjective."	
3	"With a flat prior, Bayes is ML."	
4	"Metropolis needs $p(Y)$."	
5	"A prior worth 10 means we invented 10 respondents."	
6	"A 95% confidence interval and a 95% credible interval are the same thing."	

Appendix A Pocket card

PART I Reading coefficients

GLM (L7). $Y_i \sim$ dist., $\eta_i = x_i^\top \beta$, $g(\mathrm{E}[Y_i \mid x_i]) = \eta_i$. One unit of x_{im} adds β_m to η_i. **Routine:** sign $\to e^{\hat\beta_m} \to e^{c\hat\beta_m}$ (doubling: $c = \log 2$) $\to$ predicted quantity if you need probabilities, counts or survival. Interpret $\hat\beta$, not β; ceteris paribus.

OLS (L2). $\hat\beta = (X^\top X)^{-1} X^\top y$; FWL: slope of $\tilde y$ on $\tilde x$; $\beta^\dagger = \mathrm{E}[XX^\top]^{-1}\mathrm{E}[XY]$ is the best linear predictor. Identity link: $\hat\beta_m$ is the effect.

LPM. $P(Y = 1 \mid x) = x^\top \beta$; $\hat\beta$ in probability points; can leave $[0, 1]$; heteroskedastic: use robust se.

Logit. $\partial \log\{\pi/(1-\pi)\}/\partial x_j = \beta_j$; $e^{\hat\beta_j}$ = odds ratio; $\hat\pi(x) = \mathrm{logit}^{-1}(x^\top\hat\beta) = e^z/(1+e^z)$; $\partial\hat\pi/\partial x_j = \hat\beta_j \hat\pi(1-\hat\pi)$. Contrast $\hat\tau_j(x) = \hat\pi(x_j{=}a, x_{-j}) - \hat\pi(x_j{=}b, x_{-j})$. PEA: at the average x. APE: $\frac{1}{n}\sum_i \hat\tau_i$ (prefer). Probit: $F = \Phi$; compare $\hat\pi$, not $\hat\beta$.

Uncertainty for $T(\hat\beta)$. $\hat\beta \overset{a}{\sim} N(\beta, [-H]^{-1})$; draw $\beta^{(k)} \sim N(\hat\beta, \widehat{\mathrm{Var}}(\hat\beta))$, compute $T(\beta^{(k)})$; se = sd, interval = 2.5/97.5 percentiles; or delta method.

Ordered logit (L9). $Y^* = x^\top\beta + \varepsilon$, cutpoints ζ_j (no intercept). $P(Y \le j \mid x) = \mathrm{logit}^{-1}(\zeta_j - x^\top\beta)$; $P(Y = j) = \mathrm{logit}^{-1}(\zeta_j - x^\top\beta) - \mathrm{logit}^{-1}(\zeta_{j-1} - x^\top\beta)$. $\beta_m = \partial \log \frac{P(Y>j)}{P(Y\le j)}/\partial x_m$, same for all j; $e^{\hat\beta_m}$ multiplies odds of being above any cutpoint. Middle categories: direction ambiguous. APEs sum to 0.

Multinomial (L9). $P(Y = j \mid x) = \exp(x^\top\beta_j)/\sum_k \exp(x^\top\beta_k)$, $\beta_1 = 0$. $\log\frac{P(j)}{P(1)} = x^\top\beta_j$; $e^{\hat\beta_{jm}}$ multiplies $P(j)/P(1)$. $\frac{P(j)}{P(k)} = \exp\{x^\top(\beta_j - \beta_k)\}$ (IIA). Coefficient sign $\neq$ sign of APE.

Poisson, NB (L10). $\mathrm{E}[Y \mid x] = \mu = \exp(x^\top\beta)$; $e^{\hat\beta_m}$ = IRR (ratio, not difference); gap $= \hat\mu(e^{\hat\beta_m} - 1)$. Poisson: Var $= \mu$; NB: Var $= \mu + \mu^2/\theta$. Overdispersion $\Rightarrow$ se too small: quasi-Poisson ($\phi\mu$), sandwich.

Hazard (L12). $\log\{-\log(1-h_{ij})\} = \gamma_j + x_i^\top\beta$; $e^{\hat\beta_m}$ = hazard ratio (same in every j). KM: $\hat h_j = q_j/r_j$, $\hat S(t) = \prod_{j\le t}(1-\hat h_j)$; $\hat S(t \mid x) = \prod(1-\hat h_j(x))$; $S(t \mid x_1) = S(t \mid x_0)^{e^\beta}$ when one unit apart. Censoring: independent.

Hierarchical (L14). $\beta_{0j} = \mu + u_j$, $u_j \sim N(0, \sigma_u^2)$; $\hat\beta$ hold u_j fixed. $\hat\pi_c = \mathrm{logit}^{-1}(x_c^\top\hat\beta + \hat u_j)$; MRP: $\hat\theta_j = \sum_{c\in j} s_c \hat\pi_c$. Fixed effects: nuisance; random: levels.

Penalized (L5). Ridge $\hat\beta^R_\lambda = (X^\top X + \lambda I)^{-1}X^\top y$; lasso $\lambda\|b\|_1$ gives zeros; both biased; λ by CV.

Values. $\mathrm{logit}^{-1}(z)$ at $z = 0, 1, 2, 3$: 0.50, 0.73, 0.88, 0.95; for $-z$ take $1-$ these. $e^{0.5}$=1.65, e=2.72, e^2=7.39, e^{-1}=0.37, $\log 2$=0.69.

PART II Likelihood and Bayes

Likelihood (L6). $L(\theta; Y) = p(Y \mid \theta)$: a function of θ, data fixed; *not* a distribution for θ. $\ell = \log L$, $\hat\theta = \arg\max \ell$. Score $s = \partial\ell/\partial\theta$; Hessian H; $-H$ large $\Rightarrow$ spiky, informative. Newton–Raphson: $\theta^{(t+1)} = \theta^{(t)} - H(\theta^{(t)})^{-1}s(\theta^{(t)})$. $-\ell/n$ is the average log loss; normal $Y \Rightarrow$ OLS is ML.

Bernoulli. $L = \pi^r(1-\pi)^{n-r}$; $s = \frac{r}{\pi} - \frac{n-r}{1-\pi}$; $\hat\pi = r/n$; se $= \sqrt{\hat\pi(1-\hat\pi)/n}$; Wald $\hat\pi \pm 1.96$ se.

Logit ML. $s(\beta) = \sum_i (y_i - \pi_i)x_i$; $-H = X^\top W X$, $W = \mathrm{diag}\{\pi_i(1-\pi_i)\}$; $\widehat{\mathrm{Var}}(\hat\beta) = (X^\top W X)^{-1}$; assumes the model is true and n large. Separation: MLE $= \pm\infty$.

ML uncertainty. π fixed, *data* random across repeated samples. 95% CI: the 95% describes the *procedure*.

Bayes (L13). $p(\pi \mid Y) = \dfrac{L(\pi; Y)\, p(\pi)}{p(Y)} \propto L(\pi; Y)\, p(\pi)$; $p(Y) = \int L\, p\, d\pi$ only normalizes. π still fixed; the prior and posterior describe *our uncertainty*.

Beta–Binomial. $\pi \sim \mathrm{Beta}(a, b)$, $\mathrm{E}\pi = \frac{a}{a+b}$, worth $a + b$. $\pi \mid Y \sim \mathrm{Beta}(a + r, b + n - r)$. $\mathrm{E}[\pi \mid Y] = \frac{a+b}{a+b+n}\frac{a}{a+b} + \frac{n}{a+b+n}\frac{r}{n}$. A prior matters when n is small.

Summaries. Posterior mean/median/mode; 95% credible interval: $P(\pi \in \mathrm{CrI} \mid Y) = 0.95$ given model and prior; $P(\pi > c \mid Y)$.

Priors on coefficients. $\beta_j \sim N(m_j, s_j^2)$; weak $N(0, 2.5^2)$ rules out OR $> e^5 \approx 148$. Choose: let data swamp it; check sensitivity; weak; estimate it (hierarchical, empirical Bayes: partial pooling, shrinkage toward $\hat\mu$).

Approximating a posterior. Grid (100^k evaluations). Metropolis: $\pi^* = \pi^{(t)} + \varepsilon$, $\varepsilon \sim N(0, s^2)$; ratio $= \frac{L(\pi^*)p(\pi^*)}{L(\pi^{(t)})p(\pi^{(t)})}$; accept with prob. $\min(1, \text{ratio})$, else stay; $p(Y)$ cancels. Burn in; 4 chains; $\hat R < 1.05$; Stan uses HMC. Markov chain + Monte Carlo = MCMC.

Hainan. $r = 12$, $n = 15$: $\hat\pi = 0.80$, se 0.103, CI $[0.598, 1.002]$. $\mathrm{Beta}(6, 4) \to \mathrm{Beta}(18, 7)$: mean $0.72 = 0.40(0.60) + 0.60(0.80)$, CrI $[0.53, 0.87]$, $P(\pi > 0.6 \mid Y) = 0.90$. Flat prior: $\mathrm{Beta}(13, 4)$, mode 0.80, mean 0.765.

Model comparison (L10). $LR = 2\{\ell(\text{with}) - \ell(\text{without})\} \sim \chi^2_k$ (nested only); AIC $= -2\hat\ell + 2d$; BIC $= -2\hat\ell + d\log n$.

Five parts of an interpretation: quantity and contrast; size of change; values of other x (PEA/APE); scale; uncertainty.

Sources and reproducibility. Every coefficient and table entry quoted from a lecture is the value printed on the slides (GOVT 6039, Lectures 2 and 5–14; github.com/ddekadt/cornell-govt-6039). Every derived number (predicted probabilities, odds, gaps, grid and posterior values, figures) is recomputed in `compute.py` from those printed estimates, and checked against every number the slides themselves print (61 checks). Because coefficients are printed to two decimals, a recomputed number can differ from a slide figure in the last digit. The hypothetical turnout logit (Exercise 3), the baseline survival of 0.50 (Exercise 7) and the small-province update (Exercise 11) are illustrations, not lecture data.